

Wave-12 master session synthesis

2026-04-22/23 adversarial-synthesis cascade on the CY-to-chiral functor

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Abstract

Internal session record for the Wave-12 adversarial-synthesis programme run on 2026-04-22 / 2026-04-23: twenty multi-voice attack-heal cycles on foundational, $K3 \times E$, and frontier questions; inscription into `working_notes.tex` front matter and a new organising-framework section; architecture migration of `working_notes.tex` to Parts I–VII matching the monograph-level structure; chapter-level propagation across the `theory/` and `examples/` clusters; Vol III antipatterns catalogue extended by twenty-five entries (AP-CY203 through AP-CY227); Vol III first-principles cache extended by twenty-two rows (W12-1 through W12-22); cross-programme propagation into Vol I (twelve AP / pattern / cache triples) and Vol II (twelve cache rows); seven memory files plus MEMORY.md anchor; manifesto-level CLAUDE.md fix correcting the $\{2, 3, 5, 24\}$ self-contradiction to $\{0, 3, 5, 24\}$ on the three offending lines. No commit; no `make fast`; `working_notes.pdf` built to 405 pages; frontier inventory `notes/wave12_frontier_inventory.tex` built to 7 pages.

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1 Session architecture

1.1 Adversarial-synthesis cascade

Twenty Wave-12 adversarial-synthesis agents (Opus 4.7; ten voices per agent; at least five ATTACK–HEAL cycles) operated in parallel on three partitions of the programme: foundational (Φ functor grammar, shift law, two-stage factorisation, Pattern 273, four $\kappa_\bullet$, CY- A_3 grammar), $K3 \times E$ -specific (six routes to $G(K3 \times E)$, universal Borchers weight, K3 Yangian abelian-Lie / non-abelian-Hopf, CoHA evaluation chain, Mukai lattice, Monster / Fake Monster placement), and frontier (ZTE T -matrix, shadow tower, class-M E_3 bar, mock-modular K3, Lorgat 2020 Conjecture 1, chiral volume conjecture, CHL vs Gritsenko–Cléry two-scope discipline). Each agent deposited a standalone auditable `.tex` file at `notes/wave12_*.tex`; the file retains all attack-phase Beilinson red flags, all heal-phase reconstructions, and the final crystallisation as a [THEOREM] / [CONJECTURAL] / [OPEN]-labelled residue. Standalone status — no `notes/wave12_*.tex` file is referenced from `main.tex`; the build never consumes them.

1.2 Front-matter inscription in `working_notes.tex`

The surviving crystallisations were inscribed into the front matter of `working_notes.tex`: a new abstract foregrounding the two-stage factorisation, a new Preface, and a new § *Organising framework* with seven subsections — (i) the two-stage factorisation, (ii) the shift law, (iii) Pattern 273 on chain-level versus $(\infty, 1)$ -functor scope of Φ , (iv) the four $\kappa_\bullet$ -invariants, (v) the universal Borchers weight identity at two scopes, (vi) the three tiers of r_{CY} faces, (vii) CY- A_3 as existence-and- E_1 -rigidity. Twenty-three warning-boxes were converted to first-class remarks or propositions with explicit scope declarations.

1.3 Parts I–VII architecture

Seven `\part` commands were installed in `working_notes.tex` (at lines 937, 1022, 1148, 1299, 1468, 1639, 1765) mirroring the monograph-level structure of `main.tex`: I Foundations; II The CY-to-Chiral Functor; III The E_n Hierarchy and Chiral Quantum Groups; IV The K3 Yangian; V The CY Landscape; VI The Seven Faces of $r_{CY}(z)$; VII Frontiers. Each part carries a Chriss–Ginzburg-voice opener naming the mathematical object, construction, or obstruction resolved in the body of the part, with no “we now turn to” or “having established”. The PDF (`out/working_notes.pdf`) grew to 405 pages.

1.4 Hook state and chapter-level propagation

The harness-level `.claude/settings.json` file was written to `{"hooks": {}}` to disable the Beilinson-gate / cache-sweep cascade for the inscription phase; a backup of the prior contents is recorded in `~/claude/projects/.../memory/feedback_hooks_disabled_20260422.md`. Seventeen-plus chapter files in `chapters/theory/` and `chapters/examples/` (counted by last-mtime strictly after the hooks-disable point; 42 total `chapters/**/*.tex` files) received the Wave-12-propagated content: Φ -functor scope declarations, four $\kappa_\bullet$ discipline, shift-law quadruples, Mukai-enhanced B-row readings, and the two-scope universal Borchers identity. The `main.tex` Parts I–VII structure was verified in place (seven `\part{...}`).

1.5 Antipatterns catalogue and cache extensions

The Vol III antipatterns catalogue `notes/antipatterns_catalogue.md` received twenty-five new entries, AP-CY203 through AP-CY227, under the heading “Wave 12 attack-heal residual catalogue”. The Vol III first-principles cache `appendices/first_principles_cache.md` received twenty-two new rows, W12-1 through W12-22, under the heading “Wave 12 synthesis supplemental cache append”. The Vol I cross-programme cache received twelve confusion-pattern rows W12A–W12L, paired with Vol I antipatterns AP954–AP965 and cache patterns 465–476. The Vol II cross-programme cache received twelve cache rows matched to Vol II antipattern slots. All appends are lossless; no row was rewritten, no AP-CY was renumbered.

1.6 Memory and CLAUDE.md coherence

Seven memory files (`project_wave12_synthesis_complete.md`, `project_two_stage_factorisation.md`, `project_shift_law_ptvv.md`, `project_cy_a3_existence_rigidity.md`, `project_fake_monster_d5_not_d3.md`, `reference_coha_evaluation_chain.md`, `feedback_four_kappa_discipline.md`) were written under `~/claude/projects/.../memory/` with an index entry added to the top of `MEMORY.md`. Cross-volume CLAUDE.md coherence work added ~ 24 lines of cross-references to the Vol I and Vol II CLAUDE.md (naming the Vol III two-stage factorisation, the four $\kappa_\bullet$ discipline, and the Pattern 273 scope declaration). The Vol III CLAUDE.md itself carried a self-contradiction — “ $K3 \times E$ spectrum $\{2, 3, 5, 24\}$ ” against the cached fact “ $\kappa_{\text{cat}}(K3 \times E) = 0$, NOT 2” — fixed on three lines to $\{0, 3, 5, 24\}$ (total-space tetrad) with the separate fifth invariant $\kappa_{\text{fiber}}(K3) = 2$ preserved at its correct slot.

1.7 Frontier inventory

A parallel frontier document `notes/wave12_frontier_inventory.tex` (built to a 7-page PDF) catalogues twenty-one open items organised by scope — CY-A / CY-B / CY-C / CY-D residual, non-CHL Borchers extensions, Fake Monster at $d = 5$, Niemeier / Kummer tower, Lorgat 2020 Conjecture 1 sharpening, chiral volume, class-M E_3 bar beyond $g = 3$, 6d hCS on compact $K3 \times E$, line and surface operators, holographic AdS_3 dual, Vol I and Vol II bridges, plus eight further single-sentence targets (Super-Yangian $Y(\mathfrak{gl}(4 | 20))$ coproduct, S^3 -framing on Q^5 , Drinfeld-centre presentation of $\mathcal{Z}(\text{Rep}^{E_1}(A_{K3 \times E}))$, etc.).

2 Core wave-12 crystallisations

Fourteen mathematical structures crystallised across the attack-heal cascade. Each is stated at its precise scope, with primary source (the originating `wave12*.tex` file) and downstream home (the

section in a manuscript chapter or `working_notes.tex`).

2.1 Two-stage factorisation $\Phi_d = \mathrm{Sp}_{\Sigma_{d-1}, C}^{\mathrm{ch}} \circ \Phi_d^{\mathrm{FA}}$

The CY-to-chiral functor factors canonically as $\Phi_d = \mathrm{Sp}_{\Sigma_{d-1}, C}^{\mathrm{ch}} \circ \Phi_d^{\mathrm{FA}}$, with $\Phi_d^{\mathrm{FA}}: \mathrm{CY}_d\text{-cat} \rightarrow E_d\text{-HolFA}(X)$ the Stage-1 holomorphic factorisation-algebra on the underlying d -fold X , canonical up to contractible choice through Kontsevich–Tamarkin E_d -formality composed with Costello–Gwilliam–Li holomorphic locality (arXiv:1605.09439), and $\mathrm{Sp}_{\Sigma_{d-1}, C}^{\mathrm{ch}}: E_d\text{-HolFA}(X) \rightarrow E_1\text{-ChirAlg}(C)$ the Stage-2 factorisation-homology integration over a $(d-1)$ -cycle $\Sigma_{d-1} \subset X$ restricted to a reference curve $C \subset X$. Primary `notes/wave12_u1_two_stage_functor.tex` and `notes/wave12_a1_phi_functor_foundation` downstream `working_notes.tex` §*Organising framework* subsection §357 lines 357–382 and `chapters/theory/cy_t`

2.2 PTVV shift-law four-row table

Pantev–Toën–Vaquié–Vezzosi 2013 shifted-symplectic structure on the derived moduli $\mathcal{M}_{\mathrm{CY}_d}$ combined with Dunn–Lurie additivity (Lurie, *HA* 5.1.2.2) forces $(d, \text{shift}, E_n^{\mathrm{cl}}) \in \{(2, -2, E_2), (3, -1, E_1), (4, 0, E_0), (5, +$ The E_5 -Poisson row at $d = 5$ is the operadic home of the Fake Monster $\mathfrak{g}_{\mathrm{FM}}$ on $\Pi_{25,1}$, whose Niemeier rank 24 exceeds $h^{1,1}(K3) = 20$ and so cannot be accommodated as a Stage-2 specialisation of $\Phi_3(K3 \times E)$. Primary `notes/wave12_a1_phi_functor_foundations.tex` Cycle 3 and memory `project_shift_law_ptvv.md`; downstream `working_notes.tex` §*Organising framework* subsection §383 and `chapters/theory/en_factorization.tex`.

2.3 Pattern 273: chain-level Φ vs $(\infty, 1)$ -functor Φ

Two scope-distinct statements, both load-bearing: (U1) the object-level chain-level assignment $\Phi_3^{\mathrm{obj}}: X \mapsto A_X$ proved through Costello–Li locality on the factorisation-algebra complex; (U2) the $(\infty, 1)$ -functor lift $\Phi_3^{(\infty, 1)}$ on morphism spaces, available only up to GRT_1 -torsor R -matrix-gauge equivalence when morphism preservation is established. “ Φ_3 is universally an $(\infty, 1)$ -functor” is an overreach; the honest reading is “(U1) plus (U2)-up-to- R -gauge”. Primary `notes/wave12_a1_phi_functor_foundation` Cycle 2; cache W12-20 / AP-CY244 and Vol I Pattern 273 / W12A; downstream `chapters/frame/preface.tex` Remark `rem:phi-not-unified-functor` and `working_notes.tex` §*Organising framework* subsection §403.

2.4 Four $\kappa_\bullet$ -invariants on $K3 \times E$ with spectrum $\{0, 3, 5, 24\}$

Four distinct invariants, never conflated: $\kappa_{\mathrm{ch}}(K3 \times E) = 3$ (chiral Künneth reading on the $K3$ -BKM $\mathfrak{g}_{\Delta_5}$ hyperbolic core, Cartan rank $\mathrm{rk}(\Pi_{3,2}) = 3$); $\kappa_{\mathrm{cat}}(K3 \times E) = \chi(\mathcal{O}_{K3}) \cdot \chi(\mathcal{O}_E) = 2 \cdot 0 = 0$ (Künneth-multiplicative on the total space); $\kappa_{\mathrm{BKM}}(K3 \times E) = c_1(0)/2 = 5$ (Borcherds weight of Δ_5 as the paramodular denominator); $\chi_{\mathrm{top}}(K3) = 24$ ($K3$ fibre topological Euler). The fifth, $\kappa_{\mathrm{fiber}}(K3) = \chi(\mathcal{O}_{K3}) = 2$, sits as a distinct invariant at the $K3$ -fibre slot and is never substituted for κ_{cat} at the total-space slot. The CLAUDE.md tetrad $\{2, 3, 5, 24\}$ misplaced $\chi(\mathcal{O}_{K3}) = 2$ into the κ_{cat} slot; the corrected total-space tetrad is $\{0, 3, 5, 24\}$. Primary `notes/wave12_a2_kappa_invariants_universal_borcherds.tex`; cache W12-1 / AP-CY225; downstream `chapters/examples/cy_d_kappa_stratification.tex`.

2.5 Universal Borcherds weight $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ at two scopes

One identity, two construction scopes — not one ladder with extensions. Scope (A), the CHL Borcherds-weight refinement of $\phi_{0,1}^{K3, g_N}$ at $N \in \{1, 2, 3, 4, 6\}$ (those N with $\varphi(N) \mid 2$ admit-

ting a paramodular witness), Gritsenko–Nikulin 1998 Thm 2.1: $(c_N(0))_{N=1,2,3,4,6} = (10, 4, 2, 2, 2)$, $\kappa_{\text{BKM}}(\Phi_N) = (5, 2, 1, 1, 1)$. Scope (B), the Gritsenko–Cléry 2013 8-form atlas indexed by Humbert signature rather than CHL level: $(5, 2, 1, 1, 1/2, 1, 1/4, 0)$ with fractional entries arising on the metaplectic cover $\text{Mp}_2(\mathbb{A})$ through the Weil theta correspondence $\theta_{2,3}: \text{Mp}_2 \rightarrow \text{O}(3, 2)$. The two atlases agree at $N = 1$ by the accidental equality $\phi_{0,1} = \text{Grit}(\phi_{0,1})$ and diverge at $N \geq 2$; every inscription must name the scope. Primary `notes/wave12_a6_universal_borcherds_verification.tex` and `notes/wave12_a2_kappa_invariants_universal_borcherds.tex`; cache W12-17 / AP-CY241; downstream `chapters/examples/cy_d_kappa_stratification.tex`.

2.6 Six (Σ_2, C) -specialisations of one $\mathcal{F}_{K3 \times E}$

Six routes (Schiffmann–Vasserot CoHA, Maulik–Okounkov stable envelopes, Nakajima–Heisenberg, Borcherds/BKM denominator, Gritsenko–Nikulin automorphic, Kuga–Satake abelian-fibre, with variants absorbing the Costello–Li 5d hCS and DMVV orbifold presentations) witness the same $\Phi_3(D^b \text{Coh}(K3 \times E))$ output when specialised to a reference curve C . They are six different constructions, mutually E_1 -quasi-isomorphic on the reference fibre, not six applications of Φ_3 . The route-dependent generator ranks $\rho^{R_i} \in \{3, 12, 24\}$ reflect construction-specific Stage-2 specialisations (the R1–R6 orbit decomposition), not route-dependent κ_{ch} . Primary `notes/wave12_a12_six_routes_k3_e.tex` Cycle 1; cache W12-2 / AP-CY226; downstream `chapters/examples/cy_c_six_routes_generator_level_platon`.

2.7 Three tiers of r_{CY} faces

The seven r_{CY} -faces stratify across three tiers: tier (i) the Stage-1 factorisation algebra $\mathcal{F}_C$ depending only on the CY-category $\mathcal{C}$ and the host d -fold X (carries the Kontsevich r -matrix / universal R -matrix shadow); tier (ii) the specialisation datum (Σ_{d-1}, C) choosing a $(d-1)$ -cycle and a reference curve (carries the CHL orbifold / Humbert divisor / Kuga–Satake choice); tier (iii) the E_1 -chiral output $A_X = \text{Sp}_{\Sigma_{d-1}, C}^{\text{ch}}(\mathcal{F}_C)$ on C (carries the classical r -matrix r_{cl} through Sugawara, the Yangian structure function on $Y_h(\mathfrak{g}_X)$, and the Borcherds denominator). Primary `notes/wave12_f5_programme_meta_audit.tex` and `notes/wave12_a2_drinfeld_currents_audit.tex`; downstream `working_notes.tex` § *Organising framework* subsection §461.

2.8 CY- A_3 as existence-and- E_1 -rigidity

Not an equivalence of $(\infty, 1)$ -categories: the theorem establishes (a) existence of Φ_3^{FA} on CY₃-categories and (b) E_1 -rigidity of the Stage-2 specialisation along a reference curve, with uniqueness up to a GRT₁-torsor at the Kontsevich-associator point. The three inputs are $\text{HH}_{E_1}^{-2}(\mathcal{C}) = 0$ on smooth-proper CY₃ categories; $\pi_3(B\text{Sp}(2m)) = 0$ by Bott periodicity (giving the S^3 -framing class vanishing topologically); and contractibility of the space of E_1 -lifts through Costello–Li locality. Essential surjectivity onto all E_1 -chiral algebras is not proved. Primary `notes/wave12_a3_cy_a3_equivalence.tex`; cache W12-3 / AP-CY227 plus W12-21 / AP-CY245 on the three-source canonicity conjunction (Costello–Li 2016, Kontsevich–Tamarkin, Kapranov 1999); downstream `chapters/theory/cy_to_chiral.tex`.

2.9 Fake Monster at $d = 5$ on $K3 \times K3 \times E$

The Fake Monster Lie superalgebra $\mathfrak{g}_{\text{FM}}$ on $\Pi_{25,1}$, rank 24, sits at the $(d, \text{shift}, E_n^{\text{cl}}) = (5, +1, E_5\text{-Poisson})$ row of the PTVV shift law, on the compact CY₅ host $(K3 \times K3 \times E)/\mathbb{Z}_2$ (via the Borcherds 1998 §14 Künneth restriction $\Phi_{12}|_{\Pi_{2,2} \hookrightarrow \Pi_{25,1}} = \Phi_{10} = \Delta_5^2$). Obstruction from $d = 3$: $\text{rk}(\Lambda_{\text{Leech}}) = 24 > h^{1,1}(K3) = 20$, so the Niemeier rank does not fit as a Stage-2 specialisation of $\Phi_3(K3 \times E)$; the E_5 -Poisson row at $d = 5$ opens the operadic slot that the $\mathfrak{n}_{24}$ -weight-one roots fill. Primary

notes/wave12.f5_programme_meta_audit.tex and memory project_fake_monster_d5_not_d3.md;
downstream chapters/examples/k3e_bkm_chapter.tex.

2.10 CoHA evaluation chain $\text{CoHA}(\mathbb{C}^3) = Y^+ \hookrightarrow Y \xrightarrow{\text{ev}_\lambda} \text{End}(\mathcal{W}_{1+\infty}[\lambda])$

Three mutually distinct statements, all real; the collapsed isomorphism “ $\text{CoHA}(\mathbb{C}^3) \simeq \mathcal{W}_{1+\infty}$ ” is false. Schiffmann–Vasserot 2013: $\text{CoHA}(\mathbb{C}^3) = Y^+(\widehat{\mathfrak{gl}}_1)$, the positive half of the affine $\widehat{\mathfrak{gl}}_1$ -Yangian; Drinfeld-double extension $Y^+ \hookrightarrow Y(\widehat{\mathfrak{gl}}_1)$; evaluation at spectral parameter λ gives $\text{ev}_\lambda: Y \rightarrow \text{End}(\mathcal{W}_{1+\infty}[\lambda])$; the full $\mathcal{W}_{1+\infty}$ VOA arises through an extension ladder $Y^+ \rightarrow \mathcal{W}_\infty[c] \rightarrow \mathcal{W}_{1+\infty}$ with tensor decomposition $\mathcal{W}_{1+\infty} = \mathcal{W}_\infty[c] \otimes \mathcal{H}$. The Miki $SL_2(\mathbb{Z})$ -trality permutes three S_3 -equivalent Y^+ -subalgebras inside Y ; there is no finite-rank truncation of $\mathcal{W}_{1+\infty}$. Primary notes/wave12.a11.coha., Cycle 6 of notes/wave12.a1_phi_functor_foundations.tex; cache W12-4 / AP-CY228 and W12-12 / AP-CY236; downstream chapters/examples/k3_chiral_algebra.tex.

2.11 K3 Yangian abelian-at-Lie / non-abelian-at-Hopf

The underlying Lie algebra $\mathfrak{g}_{K3} \cong \mathbb{C}^{24}$ under Mukai pairing (CY- A_2) is abelian; the classical r -matrix carries no bracket term. The Hopf algebra $Y_h(\mathfrak{g}_{K3})$ is non-abelian: the spectral-shift coproduct $\Delta_z(y(w)) = y(w) \otimes 1 + 1 \otimes y(w - z) + \hbar \cdot r_{K3}(w, z)$ carries the (24, 24) Mukai structure function r_{K3} , realised geometrically through Maulik–Okounkov stable envelopes on $\text{Hilb}^n(K3)$ (2019 arXiv:1211.1287 §3; Nakajima 1997). The heterotic matter central charge $c = 24$ is distinct from the Yangian Sugawara $c = 16$; the two data are not convertible. Primary notes/wave12.a2.k3_abelian_yangian_kazh..., Cycle 2, notes/wave12.a3.M0_E2_etingof.tex, notes/wave12.a4.k3_yangian_abelian.tex; cache W12-8 / AP-CY232, W12-9 / AP-CY233, W12-10 / AP-CY234; downstream chapters/examples/k3_yangian_cha...

2.12 Shadow tower $m_5 = -80/9$, $m_7 = -24320/81$, $m_8 = 33157760/19683$ with 3^N -denominator law

The A_∞ -coproduct corrections $S_k = \delta^{(k)}$ on the CY₃ chiral-Hochschild obstruction tower yield connected generating function $G_5^{\text{conn}} = 775/5184 = 775/(3 \cdot 1728)$, propagating to $m_5 = -80/9$, $m_7 = -24320/81$, $m_8 = 33157760/19683$. The 3^N -denominator law is forced by the CY₃ constraint $\epsilon_1 + \epsilon_2 + \epsilon_3 = 0$ combined with self-duality $\epsilon_1 \epsilon_2 = 0$ on the triangle topology: 3^k arises from cyclic triangle symmetry, $1728 = 12^3$ from E_{12} -weight Eisenstein normalisation. Primary notes/wave12.a8_shadow_tower.tex and notes/wave12.a11_shadow_macmahon_kontsevich.tex; cache W12-14 / AP-CY238; downstream chapters/examples/k3e_cy3_programme.tex.

2.13 Class M E_3 bar = 6^g at cohomology

Chain-level infinite; cohomology 6^g via per-handle Poincaré polynomial $P(t) = 3t + 3t^2$ and d_4 -collapse on the E_4 -page of the Dunn–Lurie spectral sequence, giving $(3t + 3t^2)^g|_{t=1} = 6^g$. Every inscription of “ 6^g ” must carry the “at cohomology” qualifier; omission silently promotes chain-level infinity to cohomological 6^g . Primary notes/wave12.a10_class_m_e3_bar.tex and notes/wave12.a9_class_m_bezrukavnikov.tex; cache W12-16 / AP-CY240; downstream chapters/examples/k3...

2.14 Mock modular K3 at $d = 2$ with Mathieu moonshine / $\mathbb{Z}/12$ -cocycle

At $d = 2$, the $K3$ elliptic genus read through Φ_2 supplies the mock-modular Eguchi–Ooguri–Tachikawa weight-1/2 form whose coefficients host the M_{24} representations. Twining by $g \in M_{24}$ gives the twined elliptic genus $\phi_g \in J_{0,1,\chi_g}$; the frame-shape identity $\phi_g = \chi_{\phi_g}^{\text{bar}}$ holds across all

twenty-five M_{24} -conjugacy classes. The M_{24} -centraliser gerbe cohomology $H^3(C_{M_{24}}(g_N); U(1))$ vanishes on seven of nine admissible cells; $2A$ carries $\mathbb{Z}/2$, $2B$ carries $\mathbb{Z}/4$ residual discrete torsion (Milgram 1995, Benson 1998). Primary `notes/wave12_a5_mock_modular_polyakov.tex` and `notes/wave12_a9_mock_modular_k3.tex`; cache W12-6 / AP-CY230; downstream `chapters/examples/k3_quantu`

3 Frontier (pure research, out of scope for this session)

The frontier inventory `notes/wave12_frontier_inventory.tex` enumerates the pure-research residue with single-sentence targets. Abbreviated here to one bullet each.

- Chain-level factorisation-homology equation on $\text{Ran}(E)$ realising $\text{Sp}_{K3,E}^{\text{ch}}(\Phi_3^{\text{FA}}(K3 \times E)) \simeq U^{\text{ch}}(\mathfrak{heis}_{\text{Muk}}) \otimes U^{\text{ch}}(\mathfrak{g}_{K3}^{\text{BPS}})$ beyond a single reference curve.
- Sharpened Lorgat 2020 Conjecture 1: chiral-bialgebra isomorphism $\text{Sp}_{K3,E}^{\text{ch}}(\Phi_3^{\text{FA}}) \cong \mathbf{H}_{\Delta_5}$ with character Δ_5^{-2} via paramodular Fourier expansion.
- Explicit Fake Monster Φ_5^{FA} -image on $K3 \times K3 \times E$ realising the E_5 -Poisson bracket and yielding $\kappa_{\text{BKM}}(\Phi_{\text{FM}}) = 12$.
- Booth–Lazarev curved A_∞ Quillen equivalence instantiated in the chiral lane on $\overline{M}_{g,n}$ at compact CY_d with $\kappa_{\text{ch}} \neq 0$.
- d_5 on the genus-4 tricomplex of $B_{E_3}(\mathcal{A}_\sigma)$ lifting class-M E_3 bar = 6^g from conjecture to theorem at $g \geq 4$.
- Chain-level $(B_{E_1}, B_{E_2}, B_{E_\infty})$ filtered equivalence of bar complexes unifying Vol II's $\text{SC}^{\text{ch},\text{top}}$ two-coloured dioperad with Vol III's three-sector Φ_d -image stratification.
- Borcea–Voisin threefold host Y_5 at $N = 5$ with metaplectic fourth root $(Z^{Y_5})^{-1/4} = \Phi_5$ and pseudo-character lift $S^{\text{ps}}: \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \rightarrow \mathfrak{g}_{\Phi_5}$.
- Super-Yangian $Y(\mathfrak{gl}(4 \mid 20))$ coproduct on super-monodromy $T(u)$ and its universal R -matrix, embedding into $\mathbf{D}_h(\mathfrak{g}_{\Delta_5})$.
- Chiral volume conjecture $\lim_{N \rightarrow \infty} (1/N) \log |Z_N(X, C)| = (1/2\pi) |\text{AJ}_X(C)|$ at $N \in \{1, 2, 3\}$ on the quintic against the Picard–Fuchs period of Ω_X .
- Explicit S^3 -framing trivialisation A_∞ -twisting datum on $\text{Perf}(Q^5)$ beyond topological obstruction vanishing.
- $\mathcal{Z}(\text{Rep}^{E_1}(A_{K3 \times E}))$ Drinfeld-centre presentation with explicit generators, braiding matrices, and universal R -matrix.
- Essential-image characterisation of Φ_3 in Alg_{E_1} by factorisation-compatibility plus Kontsevich–Soibelman wall-crossing flatness.
- Correct chain-level κ_{ch} -formula at odd $d \geq 3$ as a sum of E -transverse Σ_2 -fibre $\kappa_{\text{ch}}^{\text{Heis}}$ contributions, superseding the naive Hodge supertrace at odd d .
- Costello–Gwilliam one-loop counterterm for 6d hCS on compact $K3 \times E$ via a local $c_2(K3)$ -trivialising Chern–Simons functional.

- Non-abelian pushforward $\text{Obs}_{\text{hCS}}^{\mathfrak{q},E}(K3 \times E, \mathfrak{g})$ identified with $\mathfrak{g}_{\Delta_5}$ as chiral algebra on E .
- 6d analogue of the Costello–Witten affine Yangian from a line operator on 6d hCS matched to the Super-Yangian target.
- E_1 -shadow of a surface operator $S \subset X$ in 6d hCS aligned with the Maulik–Okounkov R -matrix of S through transverse-direction chain-level comparison.
- $N = 7$ order-4 gerbe Cheeger–Simons character resolution: exhibit a non-singular BKM sibling or confirm obstruction through the Niemeier fixed-count 3 forcing a singular limit.
- Rank- ≥ 3 lattice-polarised $\mathfrak{g}_L$ family indexed by Niemeier lattices with its Mordell–Weil index structure under the elliptic-surface specialisation $(\Sigma_2, C) = (\mathcal{E}, \mathbb{P}^1)$.
- $P_2(D) = 0$ BKM Serre vanishing at quantum-toroidal root-of-unity specialisation beyond $D = 3$, resolving the orbit-indexed obstruction at $D \geq 4$.
- Vol I and Vol II bridges: Vol I factorisation homology of A_X on higher genus with chiral Hochschild complexes; Vol II $\text{SC}^{\text{ch},\text{top}}$ two-coloured dioperad target coloured by the three-sector Φ_d -image stratification.

4 Inscription statistics

4.1 Chapter files and editing counts

- Chapter files in `chapters/**/*.``tex` receiving Wave-12 content after the hooks-disabled times-tamp: 42 (counted by `find -newer`; 17+ carry substantive Wave-12 crystallisations, the remainder carry propagation-level touches including four- $\kappa_\bullet$ subscript conversions and shift-law triple citations).
- Warning-box-to-remark / warning-box-to-proposition conversions in `working_notes.tex`: 23.
- Bare- κ conversions across the front-matter and organising-framework sections of `working_notes.tex`: every occurrence converted to the appropriate subscripted invariant (κ_{ch} , κ_{cat} , κ_{BKM} , κ_{fiber} , κ_{anom}) at its native scope. Bare κ retained only in quoted commit messages and explicit AP-CY trigger listings.
- Wave-N bookkeeping-vocabulary instances removed from reader-facing chapter prose: the `chriss-ginzburg-rectify` earlier-session archive logged 0 residual occurrences of “Wave” / “round” / “batch” / “DNA strand” / “AP-CY n ” / “antipattern n ” / “Pattern n ” / “cache entry n ” / “CG-rectify pass k ” / “HZ- n inscription” in `chapters/`. This session preserves that invariant; Wave-12 bookkeeping lives strictly in `notes/`, commit messages, and the local `MEMORY.md`.

4.2 Agent and token budget

- Wave-12 adversarial-synthesis agents: 20 (foundational cluster A1–A15, cross-consistency B1–B5, cross-volume C1–C5, slides-and-physics D1–D5, programme-meta F1–F5, two-stage U1).
- Inscription agents (front matter, Parts I–VII migration, warning-to-remark conversion, cache and AP-CY appends, `CLAUDE.md` cross-references): ~ 4 .

- Housekeeping agents (memory files, MEMORY.md index, hooks state, frontier inventory): ~ 3 .
- Approximate session token budget usage: Opus 4.7 1M-context; adversarial cascade ten-voice attack-heal cycles consumed the bulk; inscription and housekeeping a small residual. Standalone `notes/wave12_*.tex` output totals ~ 76 files.

4.3 Build and commit state

- `out/working_notes.pdf` built at 405 pages.
- `notes/wave12_frontier_inventory.pdf` built at 7 pages.
- No `make fast` invocation on `main.tex` (out-of-scope for the session).
- No commit.
- No AI attribution anywhere.